

Three Tabs, Mapped

One diagram per tab — the real sequence of what runs, which file it lives in, and where the paths branch or loop. Pair with *Segmentation Analysis*, *Traced* and *Trend Analysis, Traced* for the formulas behind each step.

☐ a step, tagged with its file ☐ trend-analysis path ☐ optional / looped step

TAB 1

Segmentation Analysis

A straight line, with one loop: if Auto-optimize is on, the middle three steps repeat once per parameter combination before moving on.

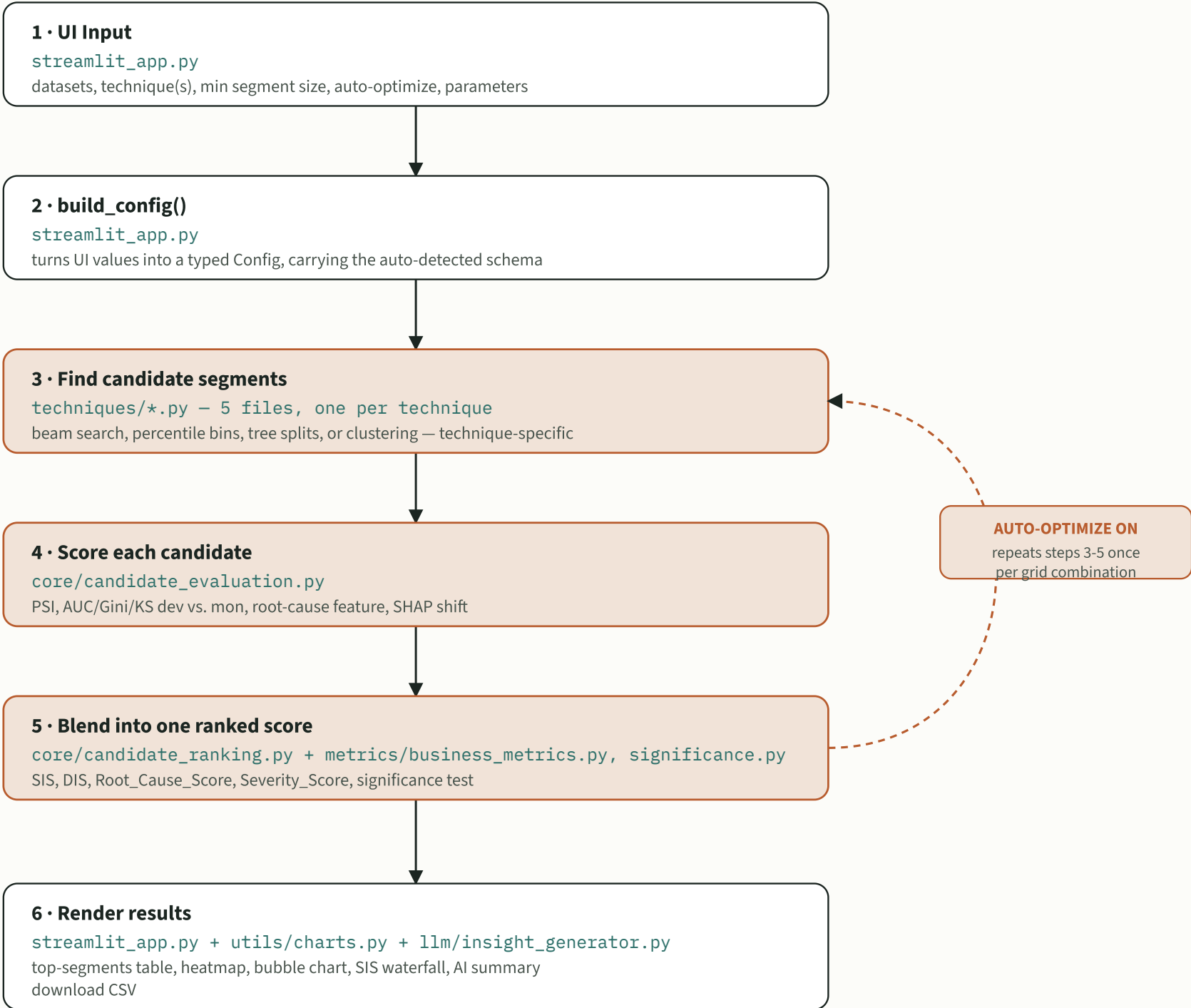


Fig. 1 — steps 3-5 (outlined) repeat once per Auto-optimize grid combination; core/parameter_optimization.py keeps whichever run scores highest by aggregate Business_Impact_Score.

- 1. `streamlit_app.py` — UI + `build_config()`
- 2. `techniques/autoslicer.py`, `feature_binning.py`, `gradient_boosting.py`, `drift_tree.py`, `kmeans.py` — candidate discovery (whichever technique(s) selected)
- 3. `core/candidate_evaluation.py` — per-candidate metrics
- 4. `core/candidate_ranking.py`, `metrics/business_metrics.py`, `metrics/significance.py` — scoring & significance
- 5. `core/parameter_optimization.py` — grid search wrapper (only if Auto-optimize is on)
- 6. `utils/charts.py`, `llm/insight_generator.py` — charts and AI summary

TAB 2

Technique Comparison

One toggle, two completely different amounts of work. Both paths run the exact Tab 1 pipeline underneath — the difference is how many times, and what gets computed on top.

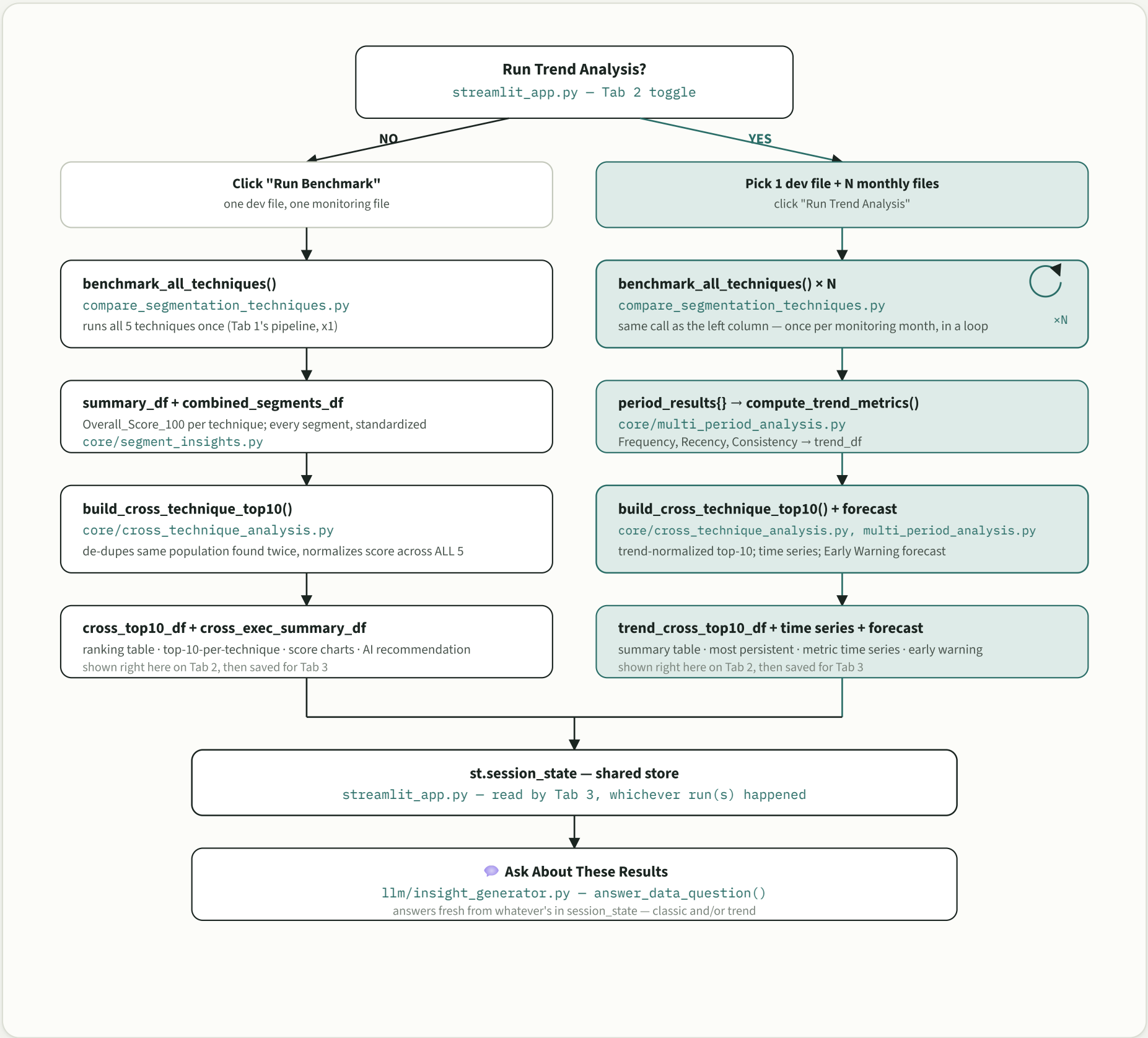


Fig. 2 — the $\times N$ loop on the right is why a trend run over 4 months takes roughly $4\times$ as long as a single classic run: it's not new logic, it's the same benchmark repeated once per month.

FILES TOUCHED, IN ORDER

- 1. `streamlit_app.py` — the toggle and both button handlers
- 2. `compare_segmentation_techniques.py` — `benchmark_all_techniques()`, called once (classic) or once per month (trend)

- 3. `core/segment_insights.py` — per-technique summary metrics, executive summary text
- 4. `core/cross_technique_analysis.py` — de-dup + true cross-technique score normalization
- 5. `core/multi_period_analysis.py` — trend path only: persistence scoring, time series, forecast
- 6. `llm/insight_generator.py` — AI Recommendation and the Q&A assistant

TAB 3

Cross-Technique Insights

The simplest of the three — nothing is computed here. Tab 3 only reads what Tab 2 already saved and displays it two ways.

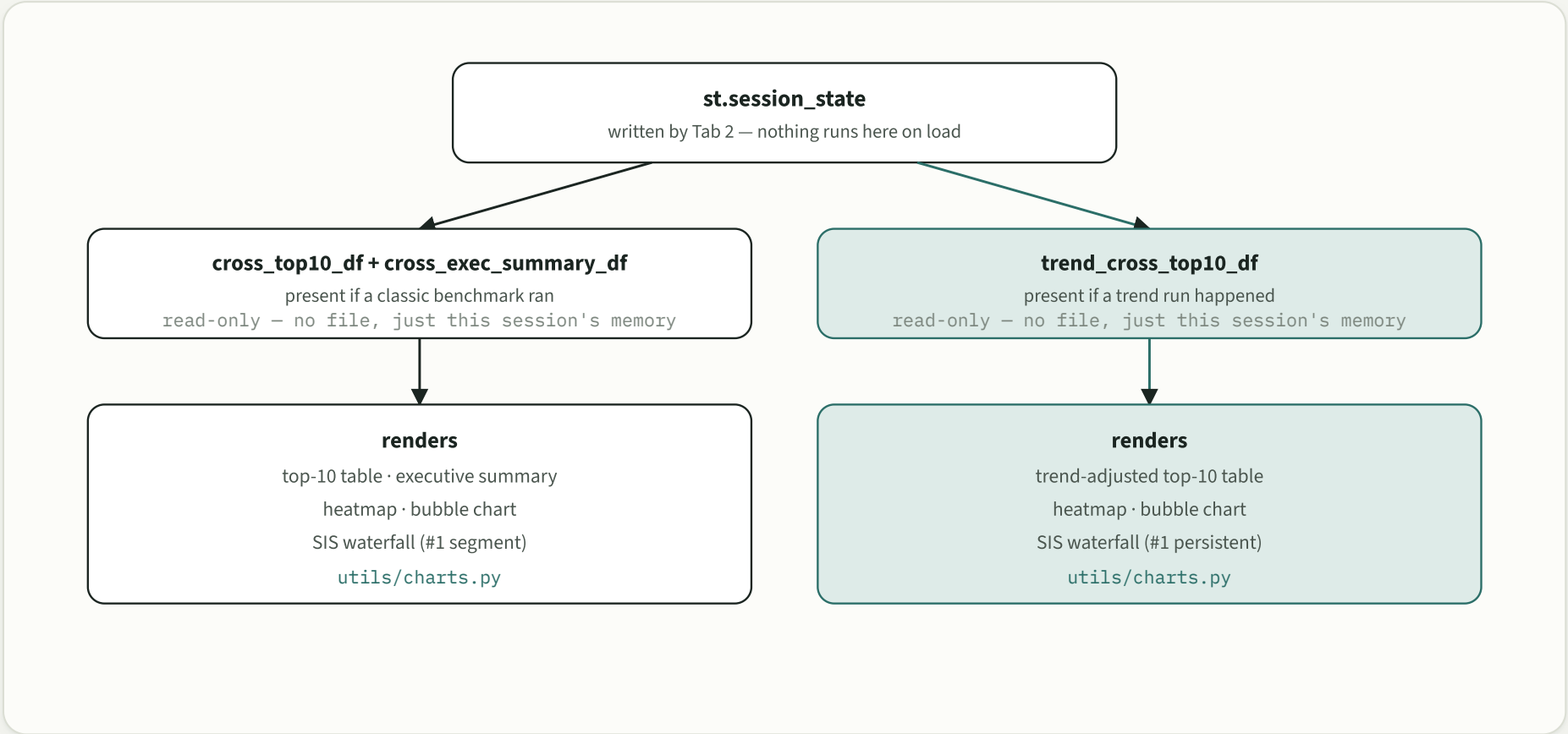


Fig. 3 — if neither run has happened yet, Tab 3 shows nothing but a prompt to go run Tab 2 first; the two sections are independent, so having only a trend run (or only a classic run) is fine.

FILES TOUCHED, IN ORDER

- 1. `streamlit_app.py` — reads `st.session_state`, no new computation
- 2. `utils/charts.py` — the three chart functions, reused identically from Tab 1 and Tab 2

Traced from `streamlit_app.py`, `compare_segmentation_techniques.py`, `core/*.py`, `metrics/*.py`, `techniques/*.py`, `utils/charts.py`, and `llm/insight_generator.py`.